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Top 16 best network monitoring tools for 2016

November 16, 2016 — by Javier Bello — 19

This post is also available in: [Spanish](#)

Towards the end of 2016 we made a short introduction to network monitoring and we told you about the main characteristics to keep in mind when selecting a network monitoring tool. This was meant for users whose installation couldn't conform with standard syslog monitoring or standard bandwidths.

To see what characteristics we talked about in order for you to make a smart choice, you can refer to that article about [network monitoring](#). In addition, read this article to get more understanding of a [network monitor](#).

Well, today we're going to choose the best network monitoring systems available now on the market (both free or licensed) and we'll compare them. Net monitoring is one of the most important sectors in network tools. What we've done on this occasion has been to use our experience with network monitoring tools, we've mounted those that were unknown to us, and we've searched other rankings and sources such as:

<https://www.slac.stanford.edu/xorg/nmtf/nmtf-tools.html>

https://en.wikipedia.org/wiki/Comparison_of_network_monitoring_systems

The objective isn't to show that Pandora FMS is the best tool in network monitoring, but instead we mean to give an objective vision so you can choose the best tool to adapt to your needs. Every installation is a world apart, and not all network tools are valid for a specific environment.

Note: all tendencies commented on this article are measured on a worldwide scale.

In depth comparison of popular network monitoring tools

Before going ahead with the review of the main network monitoring tools, we will link to our in depth comparisons related to those network monitoring tools where you can find more detailed information. We will add more reviews as soon as we get time to work on them. If you are interested on a particular comparison, please, let us know writing a comment.

[Zabbix vs Nagios vs Pandora FMS](#)

[Zenoss vs Nagios vs Pandora FMS](#)

[Cacti vs Nagios vs Pandora FMS](#)

Best network monitoring tools

Pandora FMS

The open version is capable of monitoring over 10,000 nodes and covers (without limitations) network, server (both with agents or remotely), and application monitoring. With features full of reports, alerts, and third party integrations through API, etc.

Differently from others, it doesn't have Nagios core as a starting point. Instead, it's created its on architecture from scratch, which allows perfect scaling for large environments. A network with over 10,000 nodes has been monitored with Pandora without performance issues (with the Enterprise version though).

(with the Enterprise version though).

We also highlight its integration on mobile devices, which not only allows access to the console, but also to monitoring, thanks to its geolocation system.

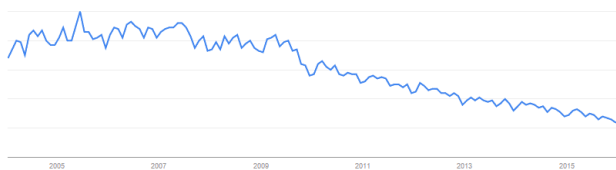
Its network autodiscovery system is capable of finding all the elements that compose your network in a short time. One of the key differentiation of Pandora FMS is that it is more than a [network monitoring](#) or [server monitoring](#) tool. It can monitor whatever you have in mind and it can scale to monitor even business processes showing the global picture of the organisation in the same tool.

You can find more information in <http://pandorafms.com/network-monitoring-tools>

Nagios

Probably the best known free tool, and it comes to no surprise since they've been working in the U.S. since 1996 to build this monitoring software. Its core is the most important part of the tool and it allows you to build plugins within that core, to monitor particular elements.

It's interesting to see how the demand tendency on the Internet has been diminishing with the passage of time. What before was one of the most potent and well-known network tools, is losing terrain.



Its large-scale use is due to the fact that it was the first one to develop a tool that covered unexpendable characteristics in a network monitoring process. For this reason, Nagios was very popular. Furthermore, given its great initial market penetration, it's still quite used.

Advantages

- There are a lot of professional profiles with Nagios experience
- If there is a good knowledge of the tool involved, manual configuration can turn Nagios into a very powerful tool to monitor isolated or particular cases
- It has a large plugin offer to adapt the product to the user's final needs
- Basic configuration is very easy

Inconveniences

- Editing or configuration processes are difficult due to the necessity to make manual modifications in order to properly set up the tool
- The GUI lacks user-friendliness.
- Steep and costly learning curve
- Every installation ends up being a "puzzle" where rather than having a standard product, we finish with a personal appliance with hundreds of patches, self-made or third-party codes. All this apart from it being complicated to evolve or maintain by said third-parties.
- Simple reports
- Very lacking when it comes to SNMP, both in polling and trap management.

Review

In a nutshell, Nagios was the origin for monitoring and, as a matter of fact, lots of new network monitoring tools have inherited the Nagios code and made it evolve. Even though you may have a lot of professional profiles on the market, these must have a very vast knowledge of the program, and your installation will depend on them 100%. Future migrations may also be complicated.

Open version: yes

Zabbix

Created by a Lithuanian company in 2005, Zabbix is known for being easy to configure and for having a very powerful GUI. It's performance starts to decline when a large quantity of nodes is to be monitored. It's agentless monitoring service stands out above the rest in its category, and experience tells us you can monitor up to 10,000 nodes without performance issues.

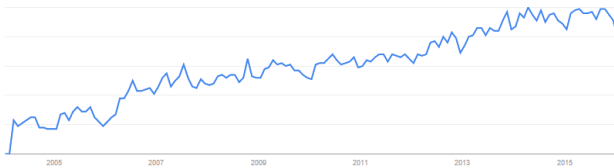
Advantages

- It has quite an active community
- At low levels, it's still very strong and efficient.

Disadvantages

- Even though it has been used for large installations, starting at 1,000 nodes its performance can be diminished
- It's difficult to create and define alert and report templates. Configurations can require many clicks and steps to be completed
- It doesn't include real-time reporting
- Difficult to purge when there are errors
- Poor SNMP trap treatment

We'll show you the interest rate graph for Zabbix:



We've got the feeling that many Nagios users are moving over to Zabbix because it's picked up on Nagios' glove and it begins to have the visibility that Nagios used to have.

Review

As we mentioned earlier, Zabbix is taking Nagios' relay and starts to appear on many installations. The problem that we can see here is with its scaling on larger CPD's. We have to tread carefully if our installation has various elements of the same type (for example databases) because their configurations will be complicated.

GroundWork

An American company that reuses different software elements from Nagios, Icinga or Cacti to create a global solution. It has managed to be among the top ranked network monitoring tools thanks to its mashup of other tools.

Review

Our experience with GroundWork hasn't been bad, but we've seen complications when integrating its different modules. Also, it doesn't have many plugins developed. For larger environments it falls short. It doesn't show an extensive history when we monitor a lot of nodes and it doesn't support platforms such as HP-UX or FreeBSD. If you don't possess a large CPD or lots of free time to tinker, we recommend checking it out since they have an interesting approach.

Zenoss

Zenoss comes from an American enterprise that has created this software to monitor storage, networks, servers, applications, and virtual servers. Its agentless monitoring is what makes it stand out. It has a "Community" version with very few features and a full-feature Commercial license.

Advantages

- All terrain regarding platforms. It's capable of cross-platform monitoring
- It presents a very flexible and tweakable dashboard, which is quite powerful as well
- Great flexibility and potential in the event management offer.

Disadvantages

- Depending on the installation's complexity and on what elements need to be monitored, adapting it can be a rough task
- The DataBase layer can get large and heavy in large environments.
- The panel can be slow on certain installations.
- It only has MySQL and its own database, lacking integration with other DataBase systems.
- Topology maps are one of the weakness of the product.

Review

Zenoss may seem interesting if you don't really want to invest in monitoring software. If you don't have that many machines and you don't plan on investing in support or large features, then you can count on Zenoss as one of your network tools. Be careful though, Zenoss version 5 requires a very potent machine to run properly.

Open version: Yes, but it may just be too limited.

Monitis

Focused toward small or medium-sized companies. It's for them that this can arise as a great network tool, and for this reason we're counting it among the 15 best network monitoring tools.

advantages

- Includes web transaction monitoring
- Allows the user to monitor common cloud applications such as Amazon or Rackspace
- Wonderful GUI which is both customizable and dynamic
- Real-time reporting

Disadvantages

- Very focused on Linux and Windows
- Has difficulties to add ad-hoc monitoring
- Doesn't have a free unlicensed version

Review

If you're a small company, this may be the tool that best suits you. We do recommend you use the fully licensed version, since the free one is kind of basic and can be easily rendered useless unless you're running a blog or similar webpage. If you're a small company, this may be the tool you need.

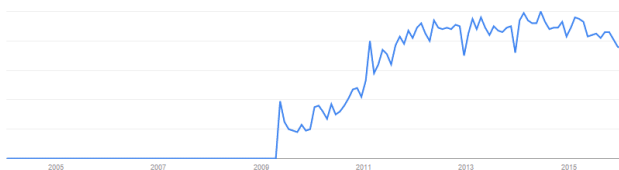
Free version: There really is none. Although, they do offer their monitor.us version, but its features are really basic and recommended for simple websites or blogs.

Icinga

Icinga comes from part of the Nagios core, over which the GUI was improved. It can be integrated with many databases and its REST API interface is outstanding for integrating other applications. It's focused on complex networks and monitoring protocols, machine resources and servers.

Review

In 2009 a Nagios fork was created, and since then it's followed its own path. In its last version (released 2014) they tried to fix performance issues by rewriting the core code. We think that it's taking the same route as Nagios. It was well received back in 2009, but after a great launch, it started to fall back in demand terms. This is yet another monitoring tool that uses the Nagios core.



Open version: Yes

Manage Engine / OPMManager

Manage Engine belongs to the Zoho Group, the giant enterprise conglomerate from India, and it's one of those monitoring softwares to

keep in mind. It's tendency regarding demand is slightly on the rise.

Advantages

- Easy to install
- Very friendly GUI
- Offers a very wide range of features to cover

Disadvantages

- Complex configurations that require a lot of documentation. Very steep learning curve.
- Complicated user experience when navigating through its screens
- Alarm levels limited to basics (warnings and critical status)
- Only available on Linux and Windows
- Lacks inventory and event correlation
- Non-existing features for large environments that are accustomed to work with events, deployment on large architectures, etc.

Review

As we mentioned earlier, it's one of those network tools that we should bear in mind a lot. What we should warn about is its really steep learning curve, which may be expensive in resources, money and time in the beginning.

Open version: Non-existent

Observium

A multiplatform web monitoring tool (mainly focused on Linux, Unix, HP).

Advantages

- Graphs on Observium are known for their amount of detail and its careful design. It's interesting for showing command panels on management levels.
- Easy and useable interface
- Capable of monitoring large installations

Disadvantages

- We weren't able to configure alerts on the free version.

Review

It's a good tool, but it lacks basic features which in some cases we can recommend complementing with Nagios or Cacti.

Open version: Yes.

Op5 Monitor

One of the monitoring tools that's more focused on hardware, network traffic, and service monitoring. It's based on Nagios as well.

It has the capability to monitor multiple platforms. It also monitors cloud systems and virtual environments. We can highlight it's capacity when it comes to large environments and its scaling on these.

From 2008 to 2010 it had a raise in demand. Ever since then, it's slow descent has begun. Its market is still Sweden, its country of origin.



Advantages

Advantages

- Easy to use
- Very good load balancing system

Disadvantages

- Hard to extend features and monitoring processes on our own.
- It won't allow deployments from the console, so they have to be done manually

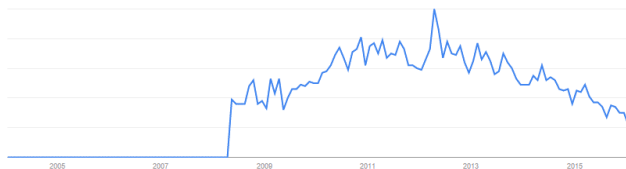
Review

Open version: Yes, they have a free version

Opsview

A tool focused on network and application monitoring. It began developing in 2003, and like many other tools seen here, it started from a version of Nagios. It has a free version available.

The demand tendency for OPSView is quite similar to that of OP5, until 2012 it has a large raise, and from then on its decline has been quick.

**Disadvantages**

- Very rigid monitoring panel
- Limited reports that cannot be exported
- The amount of developed plugins is reduced

Review

In our office we personally like this product quite a lot, even if its demand is declining. Be careful with the tool's performance.

Open version: Yes

PRTG Network Monitor

Network monitoring tool that stands out for its greatly designed and easy to use interface. It has a great vantage point when it comes to configuring alerts flexibly and because of its report generating capacities. The free version (which is NOT open) is limited to monitoring 100 application types.

PRTG is an application that can only be executed on Windows machines as a part of Microsoft Network Monitoring. Anyway, we highlight that the monitoring is multi-platform and is also able to monitor virtual systems and cloud applications. It can also generate real-time reports.

Advantages

- Very nice interface with awesome data navigation possibilities.
- You can access monitoring from mobile devices
- Reports can be in PDF/HTML formats
- Very powerful and flexible alert system

Disadvantages

- Certain plugins require additional Microsoft user licenses, therefore costs can skyrocket easily
- Limited scalability
- Very rigid when implementing its own checks

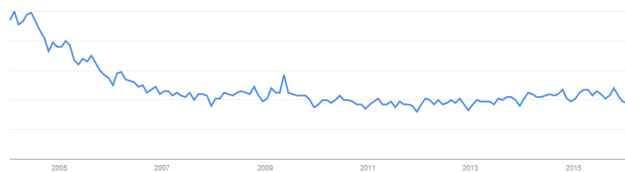
- ✓ very rigid when implementing its own checks
- Hard to deploy in environments with a complex connectivity
- Quite inefficient when it comes to server or application monitoring

Review

Open version: Yes, but very reduced

Solarwinds

Network monitoring tool that excels for its automatic network and node mapping, without the need to manually activate it. This is added to a very powerful GUI that allows you to easily view your network topology and its status. Solarwinds also allows integrating virtual machines in its monitoring.



Its tendency shows that after a crushing success experienced during 2004 and 2005, a descent happened in which they're now relatively stagnated.

Advantages

- Great GUI
- Wonderful community backing it
- Mobile device access

Disadvantages

- File and manual configuration for alerts
- The report generation system still has room to improve
- It doesn't have integrations with cloud applications such as Amazon
- You cannot perform all actions from the same panel and you'll always require to access other tools
- Its licensing forces the user to constantly look at every check performed, since licensing is individually done for these.

Review

A very good option for medium sized companies that can afford the license costs.

Open version: No.

Whatsup Gold

Whatsup Gold is one of the best network monitoring tools when it comes to balancing system loads. Its scaling is horizontal and allows the use of various processes to distribute loads. It's capable of performing some automatic actions based on events that have occurred.

Just like other tools like SolarWinds or Pandora FMS, it has a system to discover networks and topologies.

By creating proprietary scripts you can add and integrate more applications or services within the monitoring.

We can also connect from mobile devices to access the monitoring panel. Alerts can be set to be sent via email, SMS or through other options.

Information is offered in real time.





Advantages

- Easy to configure and process network discoveries
- Reports

Disadvantages

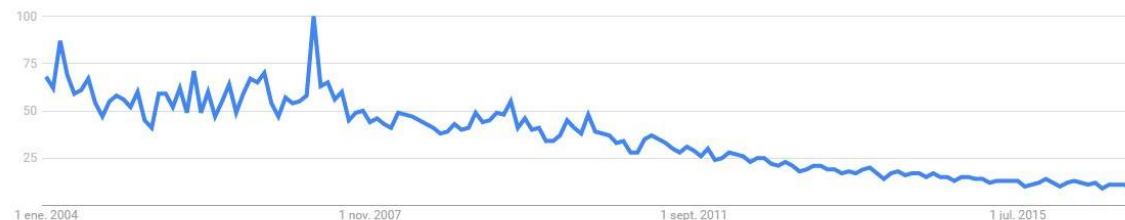
- Navigation and console are not that intuitive
- Configuration is split between web configurations and manual configurations on the console
- Limited scalability
- Quite poor for monitoring servers or applications

Review

Open version: No.

OpenNMS

A 100% open source network monitoring tool, with no licensing requirements. The business model is to offer the software free and monetize it through consulting services, provided by the OpenNMS group, who administrate the product, through various business lines, including training and support services.



As one can see in the graph, demand for OpenNMS has been on a steady decline for the last decade.

Advantages

- It's a flexible and high-capacity network monitoring tool, but one of its chief problems is the steep learning curve associated with it, plus an unintuitive GUI.
- Integrates well with alert systems supports like SMS and email.
- An active community around the tool, and a detailed wiki.
- Can perform similarly to other, licensed products, but requires more time to learn to use and to configure

Disadvantages

- Requires a "hands-on" style of configuration, based on script editing and XMLs. In this aspect, it resembles Nagios a lot, in that a manual configuration can lead to highly idiosyncratic installations which can be difficult to decipher.
- Importing third-party MIBs can be problematic.
- The reports it is capable of generating are not polished enough for executive/management eyes, being quite basic visually, and without the possibility for fine-tuning intervals. In fact, it can't present information in real-time, nor fine-tune intervals to below 24 hours.
- Most importantly, OpenNMS, as its name suggests, is focused on network monitoring, and does not cover the full range of services which a more complete monitoring service can provide, such as server or application monitoring, and much less business processes and user experience. Due to its lack of agents, the information it can retrieve from the system under monitorization is limited to what is retrievable

experience due to its lack of agents, the information can be retrieved from the system since the monitoring is limited to what is accessible through SNMP, and that is not enough when it comes to monitoring applications or elements of server infrastructure. In some cases (not our own) it has been possible to use Nagios agents in an OpenNMS integration, although we can't consider that an optimal solution.

Open version: Yes.

Microsoft Network Monitor /Microsoft Message Analyzer

Microsoft Network Monitor was discontinued and is now named Microsoft Message Analyzer. Its main feature is to analyze protocols and to have vision on network traffic. We've introduced this tool in this report because we now a lot of people search for it and ask us about it. We must know, though, that it cannot compete feature-wise with any of the formerly mentioned tools.

Conclusions

We hope this comparison of network monitoring tools has been useful. As we understand it, we've carefully selected the best market options for network tools. We know that there's a lot of competition and many options, and it is honestly quite difficult to choose the best option.

Most likely, you'll find the option you need on this list, most network monitoring tools that we mention are meant for small and medium enterprises. It's difficult to find tools for large companies that actually cut it for them in performance, and that also support different technologies and protocols that this type of company needs. In this category we can mention ZenOSS and Pandora FMS.

You should also take into account that this article discusses network monitoring tools and, today, it's more important not only to know the status of networks and applications, but to be able to understand how a business works, from the bit that goes along a network cable, to sales being done at a specific time by a specific company. This point is related to business activity monitoring, or operational intelligence; but that's better left for another article.

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19 comments



Mark

August 11, 2016 at 8:43 am

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14



We actually tried several tools from the above list. Finally we found CloudView NMS. I am surprised it is not in the list ... They say, 10K nodes is not a problem for CloudView. We have ~1,600 TCP/IP and SNMP nodes so far. Scalability is tightly connected to default polling intervals. Their default is 10 sec but after 1000 nodes we had to change it to 60sec. Looks like it can monitor anything based on standards . Optional agents running on any OS platform. More about scalability: it allows to configure a chain of connected slave-NMS servers which a report to master NMS in"Central Office". The GUI is very intuitive. It is available both via HTML-5 web interface of full-blown windows interface from any client OS (windows/linux/mac)

101

Rate this

REPLY



BigDave

September 22, 2016 at 5:03 pm

Cloudview does not really look like an enterprise monitoring system, looks like its been knocked up as a school project

18

Rate this

REPLY



Steve

October 8, 2016 at 11:42 pm

It looks like above negative comment came from a jealous competitor. CloudView NMS is used by major enterprises around the world and NASA. Security corresponds to FIPS requirement, I just suggest to try it yourself. BTW, thank the authors for the software list

the authors for the software list

30

Rate this

REPLY



littledave

November 29, 2016 at 7:02 pm

Yeah I'm with bigdave. Cloudnms gui is ugly like their website.

01

Rate this

REPLY



Victor

December 2, 2016 at 7:22 am

Well, "Big Dave" actually indicated his website. We contacted the the owner just to confirm that nobody of his small company actually left the bashing comment. As one of our customers indicated below , CloudView NMS is enterprise software without enterprise price. And yes, the website is not spectacular, so what? Most of those who run the free trial stayed with us. Disclaimer: I work in CloudView NMS

10

Rate this



Arvin

September 29, 2016 at 12:25 pm

19
Shares

Great blog. thank you sharing.



00



Rate this



REPLY



14



Arvin

September 30, 2016 at 8:34 am

Great List. Thank you for sharing.

00

Rate this

REPLY



JD

October 2, 2016 at 6:01 am

What, no Sensu, Prometheus, TICK (Influx), other modern tools?

10

Rate this

REPLY



Eddie

October 9, 2016 at 12:36 am

We are happy Cloudview customers. It is really amazing: we dropped several MUCH more expensive alternatives after we found them. The only thing which is not impressive is their website which is not exactly shining... I guess they just do not care about the website. It is true when they say "the best things in life are free". I never saw a monitoring system which can handle more than 1000 nodes from a crappy old laptop... This one can.

40

Rate this

REPLY



Gerd

October 24, 2016 at 7:22 pm

Your list misses opennms.

20

Rate this

REPLY



Javier

October 27, 2016 at 4:02 pm

Thanks for share it Gerd

00

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19
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14



Mark

October 28, 2016 at 11:16 pm

Please keep in mind that IpSwitch WhatsUpGold has gone through a complete rewrite to version 2017 and most of its abilities have changed or are not in place yet (as of SP1)

For example, the Windows app is gone, the mobile version is gone, many people are finding the new web UI dreadfully slow, etc. There are some new features but 2017 is essentially a different product now.

00

Rate this

REPLY



Javier

November 3, 2016 at 6:20 pm

Thanks Mark, we will check again the new version of IpSwitch in the neares future. Stay tuned.

00

Rate this

REPLY



Ken Flott

October 29, 2016 at 2:36 pm

NetBrain is not marketed as a monitoring tool but has excellent monitoring capabilities built in that are actionable right from the map.

10

Rate this

REPLY



Javier

November 3, 2016 at 6:18 pm

Thanks for the update Ken, sadly we do not think NetBrain can fit the list of the best 16 network monitoring tools.

However, we will have a look at your solution in case we can add it in the coming future.

Thanks for your contribution

Thanks for your contribution

10
Rate this

REPLY



Daniel van Dorp
October 29, 2016 at 8:53 pm



LibreNMS (Observium fork) should also be included, it does do alerts as there is only a free version :)

10
Rate this

REPLY



Kimberley Trommler
November 3, 2016 at 1:27 pm

Thanks for a really useful overview, Javier, and also thanks for including PRTG.

Could I make a small correction to what you wrote about PRTG? The freeware version was increased to 100 sensors in mid-2015, so you can monitor 100 things for free now, not only 10.

Thanks!

Kimberley (disclaimer: I work at Paessler!)

00
Rate this

REPLY



Javier
November 3, 2016 at 6:15 pm

THanks for this input Kimberley, we have updated to 100 things to monitor for free.

00
Rate this

REPLY



Jared
November 15, 2016 at 8:57 am

Great article, thanks! But i'm curious why there are no TNI pc audit software (Total Network Inventory) in this list? This list doesn't seems to compare tools that were made in 2016 or something like that. There are plenty of software made earlier. IMO it's great and need to be included. Also there are no Lansweeper or Spiceworks. Really? Some of the most popular free software out there?

00
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-Jonah Kowall Research Vice President, Gartner

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